

PH 602: Quantum Theory of Solids

Course Code: PH 602

Course Name: Quantum Theory of Solids

L-T-P-C: 3-0-0-3

Approval: One-time approval (OTA) recommended by the 26th BoA and approved in the 18th Senate

Course Contents

The course starts from electrons in periodic systems and works through the band structure in solids (including graphene and CNT). Electrical transport in solids is worked out from Boltzmann and Kubo formalism. The course finally works out the complex quantum organisations like magnetism and superconductivity.

- **Electrons in solids:** (12 lectures)
 - i. Electronic degrees of freedom: from a single atom to N atoms.
 - ii. “Free” electron description – why should it ever work?
 - iii. “Independent” electron description – existence of a Fermi surface.
 - iv. Electron bands: metal, semiconductor and insulator.
 - v. Quantum well, dot, wire, nanotube.
 - vi. Optical properties: nearly free electron response (Drude), optical absorption, polaritons, photoemission.
 - vii. “Quantum liquid: Landau’s Fermi liquid” – concept of an emergent quasiparticle.**
- **Electron Transport:** (6 lectures)
 - i. Electrons in a field.
 - ii. Boltzmann transport.
 - iii. Quantum of conductance, Coulomb blockade.
 - iv. Meaning of Ohm’s law, coherent transport, Landauer approach.
 - v. From atoms to quantum devices.
- **Ordering of electronic degrees of freedom: Applications:** (8 lectures)

Magnetism: A macroscopic quantum phenomenon

 - i. Magnetic coupling of electrons: Para, Ferro and Diamagnets.
 - ii. No classical analogue: Bohr van Leeuwen theorem
 - iii. Magnetic interactions: long range order, magnetic excitations
 - iv. Spintronics applications: using itinerant electron spin for transport – a new paradigm, new electronic materials, GMR and CMR
- **Introductory Superconductivity**:** (6 lectures)
 - i. Basic phenomena, Meissner effect, London equation.
 - ii. Towards a pairing mechanism: Cooper problem, BCS theory, experimental verification.
- **Two dimensional electron gas:** (6 lectures)



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- i. What is a 2D system? Instabilities of a 2D charged system.
 - ii. MOSFET configuration: 2D electron confinement.
 - iii. Electrons in a magnetic field: Landau levels.
 - iv. Hall effect: the quantized version.
 - v. Fractional quantum Hall effect: fractional quantum numbers in 2D electron gas**